

brightness greater than or equal to 1,000 nits, it is far more enhanced than the standard definition range (SDR) televisions. As video technology will progress with HDR10+ and new standards, so would the need for Smart TVs that can deliver that experience.

The second important area of innovation would be display and screen experience. Not just TVs, but even cinema and smartphone screens are constantly innovating to maximise resolution they can support. Lifelike and vivid images are created with unique pixels that can be displayed horizontally and vertically. Currently, widely available ranges are 720p HD and 1080p HD with ultra-high definition (UHD TV) or 4k resolution gaining popularity. But in the next twenty years, I foresee 8K technology being developed and rolled out. Experts agree that 8K resolution is the closest to what the human eye perceives, and hence video recording to distributing platforms will adopt the technology for the best customer experience. The 8K television sets will cost more initially, and as the technology develops further, prices will reduce. Coupled with this, full-array LED backlights will become an industry standard. We have already moved from edge lighting to direct-lit screens as a common practice today, but innovation doesn't cease here. Full-array backlighting is essentially individual control of the brightness of each LED across the back of the screen. It creates a better picture quality for customer delight. Currently, this technology is premium and the TVs look bulkier. But as the technology develops further, the full-array backlight will be an industry standard.

At this pace, Smart TVs will outpace laptops, computers and even cinema screens through innovation in both software and hardware. The aim would be to constantly improve resolutions and streaming qualities that challenge

the limitations of current viewing experiences.

The entire ecosystem in India is ready to encourage and promote Smart TVs but to harness this potential, India needs to transform the TV manufacturing industry, driven by technology and policy support.

Make in India

With the rising Make in India sentiment, Indian TV manufacturers have received the required push, but challenges are aplenty.

India is dependent on China and other south Asian countries for either import of TV sets or components. According to the commerce ministry, in 2019-20, India imported USD780.84 million TVs, of which Vietnam contributed around USD428 million while China catered to roughly USD 292 million. Despite home-grown brands, these countries cater to 36 per cent of the USD 2 billion Indian TV industry.

The challenges are compounded as India does not have the required infrastructure to manufacture a complete TV set or any electronic product locally. The most expensive component in a television set is the open cell, but it is not manufactured in India by any company. Globally, there are only 6-7 companies, mostly in China and Vietnam manufacturing open cells for decades. Hence, firstly, to be completely self-reliant, India needs to have a policy that encourages investment in such manufacturing capabilities. Even public-private partnerships or Government campaigns, such as Invest in India, should focus on developing these manufacturing capabilities.

The dependency on China is not just specific to India but is a global challenge. Some countries have concentrated their capabilities as manufacturing hubs, while some, like India, developed their services and IT. But now is the time for India to widen its focus on manufacturing, especially for

SMBs. Importing technology and components is also not sustainable for Indian companies due to externalities that increase production cost significantly. For example, in 2020, during the pandemic, international trade was restricted, and raw material prices increased by 350 per cent.

Last year in July, commerce minister Piyush Goyal announced that the government would build self-reliance in televisions, but the journey ahead is long. India needs manufacturing and technological transformation to be truly self-reliant.

Some steps in this regard have been taken. The government exempted duties on select TV components such as open cells, chips on films and printed circuit boards assembly (PCBA) but at the same time imposed a 5 per cent customs duty on the import of open-cell for televisions. Coupled with the increasing prices of raw materials, the manufacturing cost of TVs in India is increasing rapidly.

The government should strengthen the incentives like the 4-6 per cent incentive that the PLI scheme offers to the domestic TV manufacturers to help balance the high input cost. Similarly, the government can boost local production and manufacturing by introducing incentives for investments by local manufacturers and SMBs. The government can also drive demand by reducing GST on TVs from 28 per cent to 18 per cent to make TVs more accessible and affordable.

At the same time, Make in India should become the required platform to encourage more investments in TV manufacturing and technological development in the country.

In 2020 and 2021, many governments have mandated stay-at-home orders with lockdowns and curfews, giving the TV companies, in a way, a captive market. The future of Smart TVs over the next 20 years is promising. An understanding of customers, their needs and bringing them the best technology at affordable prices is crucial. 